higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

SYSTEM ANALYSIS AND DESIGN
NQF LEVEL 4

13 NOVEMBER 2019

This marking guideline consists of 12 pages.

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SECTION A

QUESTION 1

1.1	1.1.1	C		
	1.1.2	A		
	1.1.3	B		
	1.1.4	B		
	1.1.5	A		
	1.1.6	C		
	1.1.7	B		
	1.1.8	D		
	1.1.9	A		
	1.1.10	C		
			(10 × 1)	(10)
1.2	1.2.1	True		
	1.2.2	True		
	1.2.3	False		
	1.2.4	False		
	1.2.5	True		
	1.2.6	False		
	1.2.7	False		
	1.2.8	True		
	1.2.9	False		
	1.2.10	True		
			(10 × 1)	(10)
1.3	1.3.1	E		
	1.3.2	I		
	1.3.3	Q		
	1.3.4	B		
	1.3.5	M		
	1.3.6	J		
	1.3.7	A		
	1.3.8	C		
	1.3.9	P		
	1.3.10	K		
			(10 × 1)	(10)
1.4	1.4.1	Virtual reality		
	1.4.2	hardware		
	1.4.3	intranet		
	1.4.4	data-flow		
	1.4.5	open source		
	1.4.6	user-requirement		
	1.4.7	closed		
	1.4.8	Spyware		
	1.4.9	qualitative		
	1.4.10	sensor		
			(10 × 1)	(10)
				[40]

TOTAL SECTION A: 40

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QUESTION 2

2.1

- **Accountability**
  - A professional is expected to take responsibility for tasks in job functions.
  - A professional is expected to admit mistakes when things go wrong.

- **Good communication skills**
  - A professional is expected to have good skills, such as reading, writing, speaking and listening.
  - A professional is expected to have good interpersonal skills so that other people can understand him/her.

- **Integrity**
  - A professional is expected to be trustworthy and honest.
  - A professional is expected to have strong moral and ethical principles.

- **Team-working skills**
  - A professional is expected to have team-working skills to work with other people.
  - A professional is expected to cooperate, listen, trust and respect other members.

- **Ongoing professional development**
  - A professional is expected to continuously keep up to date with new and emerging technologies within the IT industry.
  - A professional is expected to continuously update skills and qualifications.

Mark allocation
1 Mark for each quality and 1 mark for an applicable fact per description.
(5 × (1 + 1)) (10)

2.2 2.2.1 A skill or behaviour that goes beyond what an ordinary person would have or behaving in a more formal or business-like manner

2.2.2 Using hiring policies that encourage fair representation of members of all race groups, women and people with disabilities

(2 × 2) (4)
[14]

QUESTION 3

3.1 3.1.1

- Malware could damage or disable the web application.
- Accidental file deletion of important web-application files could cause the web portal to malfunction.
- Hacking could lead to data/information being stolen.
- Cyber-attacks could take the website offline.
- Web-application vulnerabilities could exist in the software and technologies used.
- Data-security misconfiguration of the web-application server (4)

and database server could lead to exploitations. (Any 4 × 1)

3.1.2
- Malware – Antivirus or antimalware programs should be installed and regularly updated.
- Accidental file deletion – Correct setup of user accounts with limited/restricted security privileges.
- Hacking – Install web-security software or web-filtering products.
- Cyber-attacks –
  - Network infrastructure should be configured to prevent cyber-attacks.
  - Web servers and software should be protected by network and web-application firewalls.
- Web-application vulnerabilities – Web applications should use the latest web-application software which should be regularly updated and patched.
- Data-security misconfiguration – Those tasked with managing the web-application portal should be properly trained and skilled to avoid security misconfigurations. (Any 4 × 1) (4)

3.1.3
- Lack of user involvement
- Technological problems
- Lack of expertise
- Lack of understanding of the problem, opportunity or directive
- Incomplete specification of requirements and functionality
- Changing requirements
- Lack of management support for the project
- Lack of control or planning (Any 3 × 1) (3)

3.1.4 The expected impact of failure can be calculated as:

Risk = consequence × probability
= R750 000 × 15%✓
= R112 500✓ (2)

3.1.5 The probability of success is 100% – 15% = 85% (1)

3.1.6 Expected impact = R750 000 × 85%✓
= R637 500✓ (2)

3.1.7
- Determine the acceptable level of risk.
- Assess the likely risks to determine the actual expected risk.
- Plan to reduce the individual risk until the total expected risk falls within the accepted risk. (3)

3.1.8
- It is the most effective way to achieve data security.
- It securely protects users' passwords. (2)

3.2 Risk assessment is the measurement of the potential impact of risk by combining its consequences with its probability,✓✓ whereas risk management is the process of identifying vulnerabilities and threats to the information resources used by an organisation.✓✓ **OR** It is the process of identifying, assessing and controlling threats to an organisation's capital and earnings.✓✓ (2 + 2) (4)

3.3
- Identification
- Authorisation
- Authentication (3)

3.4
- It should be completed in time.
- It should be completed within the budget.
- It should meet the requirements and do what it is supposed to do.
- It should meet the required standards. (4)

[32]

QUESTION 4

4.1 4.1.1 Asking follow-up questions when a response is not fully understood, when answers are vague or ambiguous or when we want to obtain more specific or in-depth information

4.1.2 Feeling of trust, confidence and understanding between two people

4.1.3 A type of question that is vague or generalised and difficult for the respondent to interpret

(3 × 2) (6)

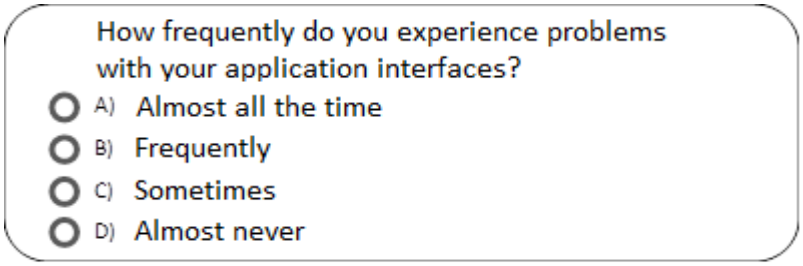
4.2

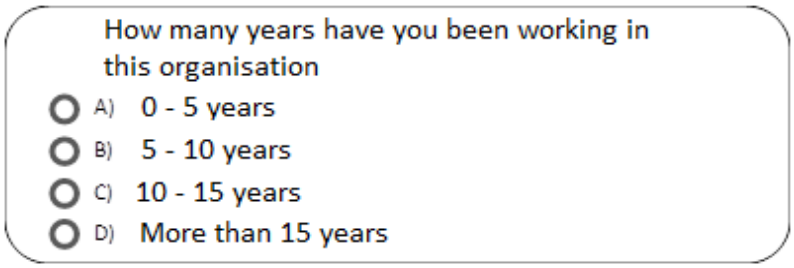
METHOD	ADVANTAGE	DISADVANTAGE
Document review	• Relatively inexpensive • May show concerns not noted by other methods	• Information may not be applicable or out of date • Time-consuming method
Observation	• Data can be collected when an event or activity is occurring. • Directly see what people are doing	• Observer can be biased • People can behave differently when they know they are observed
Interview	• Allows respondents to describe what is important to them • Useful for gaining more insight into a topic/problem	• Interview bias • Time consuming and expensive

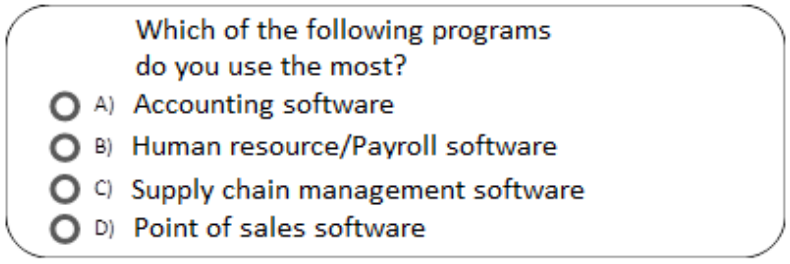
(Any relevant 3 + any relevant 3) (6)

4.3
- The time span to complete an online survey is much less than with the traditional method.
- Using an online survey reduces cost as usage or resources needed are greatly reduced.
- Online surveys are more accurate as opposed to the traditional method in which human error can occur.
- Online surveys are much faster to analyse, and so results can be viewed in real time.
- It is easier for participants as the survey can be completed as and when time is available for the participant. (Any 4 × 1) (4)

4.4 Examples of multiple-choice questions:







(1 mark for relevant question + 1 mark for correct construct of the presentation of the three questions)

(2 × 2) (4)

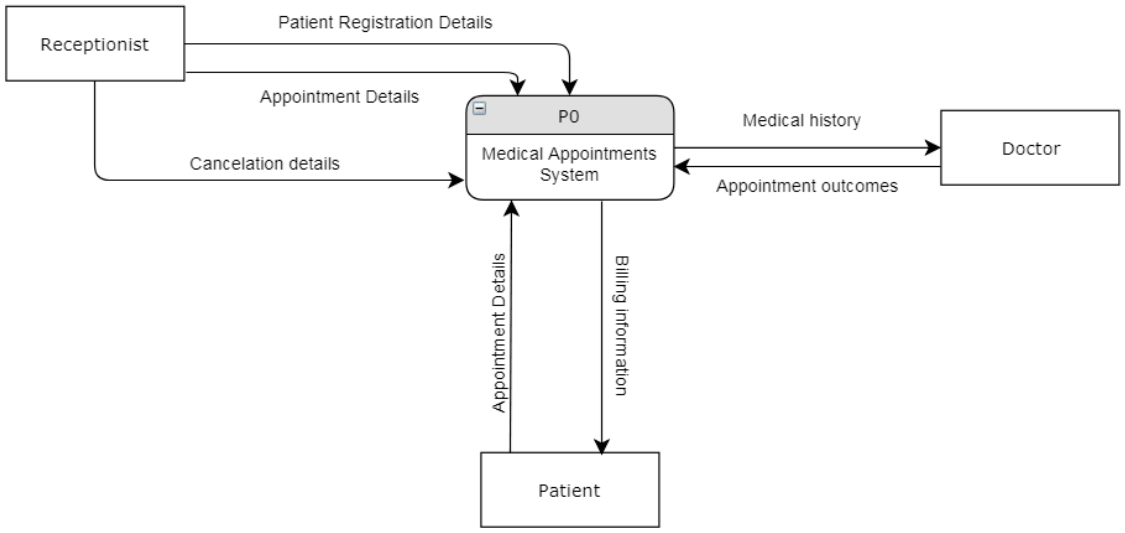
4.5 Examples of rating-scale questions:

			Strongly disagree 1	2	3	4	Strongly agree 5
1	Overall, I am satisfied with how easy it is to use this system.		O	O	O	O	O
2	It was simple to use this system.		O	O	O	O	O
3	I was able to complete the tasks and scenarios quickly using this system.		O	O	O	O	O
4	I felt comfortable using this system.		O	O	O	O	O
5	It was easy to learn to use this system.		O	O	O	O	O
6	I believe I could become productive quickly using this system.		O	O	O	O	O
7	The system gave error messages that clearly told me how to fix problems.		O	O	O	O	O
8	Whenever I made a mistake using the system, I could recover easily and quickly.		O	O	O	O	O
9	The information (such as online help, on-screen messages and other documentation) provided with this system was clear.		O	O	O	O	O

(1 mark for relevant question + 1 mark for correct construct of the presentation of the three questions)
(2 × 2) (4)
[24]

QUESTION 5

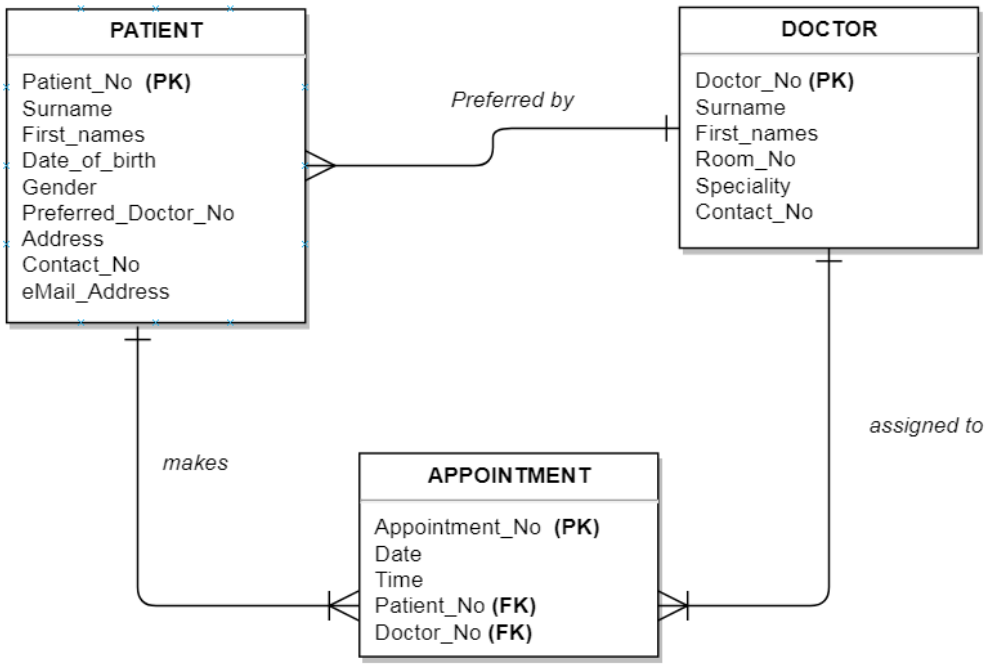
5.1



Mark allocation:

Main process (name)	1	
Indicating it is a P0 or Level 0 diagram	1	
Each external entity	3	
Each data flow with direction	6	(Maximum 6) (11)

5.2



NOTE: Complete ERD included for reference purposes.

5.2.1 APPOINTMENT (1)

5.2.2 Patient_No (FK) (1)

(Or another name referencing the same PK field) FK must be indicated as instructed

5.2.3 Doctor_No (FK) (1)

(Or another name referencing the same PK field) FK must be indicated as instructed

5.2.4 Makes (1)

Or any relevant verb

5.2.5 Assigned to (1)

Or any relevant verb

5.2.6 (1)

Indidating 1

5.2.7 (1)

or

Indidating Many

5.2.8  (1)

Indidating 1

5.2.9  (1)

or



Indidating Many

[20]

QUESTION 6

6.1 6.1.1 Yes (1)

6.1.2
- It is well organised.
- All screen objects are labelled.
- The interface is simple and uncluttered. (Any 1 × 1) (1)

6.1.3 Drop-down box/Combo box (1)

6.1.4 The drop-down box provides a list of options from which to choose. (1)

6.1.5 Radio buttons (1)

6.1.6
- Increases understanding between the designer and the user
- Reduces the cost of developing the system
- Helps the user to accept the system
- Evaluates the design to ensure usability (Any 3 × 1) (3)

6.1.7
- Ensures that a design can work in the user's place
- Gets an idea of how well the final solution will work (2 × 1) (2)

6.1.8 Text
or
String (1)

6.1.9 Preferred_Doctor_No (1)

6.1.10 1:M or M:1
1 to Many or Many to 1 (1)

6.1.11 The Age field could be calculated/derived from the date of birth (1)

6.1.12
- The email address should contain at least 1 @ symbol
- The domain should be valid
- The email address should not contain any spaces
- Alphanumeric characters and an _ (underscore), . (period) or – hyphen may be used before the @
- Special characters should be avoided

(Any acceptable 2)

(2 × 1) (2)

6.2 Online forms guide the user through the data-entry process and are flexible enough to adapt to information that a user enters, whereas offline forms are not flexible and cannot adapt to the respondent. (2 × 1) (2)

[18]

QUESTION 7

7.1 7.1.1
- Verifying that the system functions according to the system-requirement specification
- Detecting defects and flaws in the system
- Measuring the quality of the system
- Ensuring the system meets user requirements (Any 2 × 1) (2)

7.1.2
- Unit testing
- Integration testing
- User-acceptance testing (3)

7.1.3 Typical end users of the system (1)

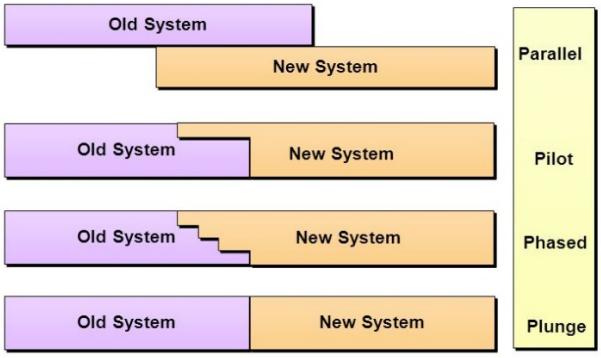
7.1.4
- Stress testing
- Load testing (2)

7.1.5
- Determining the robustness of the system
- Testing system availability under peak conditions
- Checking error handling under load
- Breaking system under test to determine if it can recover from major crashes (4)

7.2 Data specifically identified for use in testing a computer program, where the output produced are also verified. (2)

7.3 Result of a test or process that is different from what was expected (2)

7.4
- Immediate/Big Bang /Plunge
- Parallel
- Phased
- Pilot



(Source: John Wiley & Sons, Inc. 2007)

Allocate 1 mark for each diagram and 1 mark for the correct name (4 × 2) (8)

7.5
- Checking if the system has met all its objectives and satisfied all user requirements
- Checking if the system is working as efficiently as it should
- Checking if there are any lessons learnt from the design and development of this system that will benefit future systems
  and
- Accuracy, completeness, and timeliness of information system output
- User satisfaction
- System reliability and maintainability
- Adequacy of system controls and security measures
- Hardware efficiency and platform performance
- Effectiveness of database implementation
- Performance of the IT team
- Completeness and quality of documentation
- Quality and effectiveness of training
- Accuracy of cost-benefit estimates and development schedules

(Source: Shelly, G.B., Rosenblatt, H.J., 2011 Systems Analysis and Design, Course Technology Press, Boston, MA, 2011) (6 × 1) (6)

7.6
- Interview members of management and key users
- Observe users and computer operations personnel actually working with the new information system
- Read all documentation and training materials
- Examine all source documents, output reports, and screen displays
- Use questionnaires to gather information and opinions from a large number of users
- Analyse maintenance and help desk logs

(Source: Shelly, G.B., Rosenblatt, H.J., 2011 Systems Analysis and Design, Course Technology Press, Boston, MA, 2011) (3 × 1) (3)

7.7 Between 6 and 18 months after completion of the project (1)

7.8
- Corrective
- Preventative
- Adaptive
- Perfective (Any 2 × 1) (2)

[36]

QUESTION 8

8.1 8.1.1 It is a machine capable of carrying out a complex series of actions automatically, especially one programmable by a computer.

8.1.2 It is a logic-operations method based on many-valued logic rather than binary logic (two-valued logic).

OR

Type of reasoning based on the recognition that logical statements are not only true or false.

(2 × 2) (4)

8.2
- Fixed neural networks
- Adaptive neural networks (2)

8.3
- Repetitive tasks
- Dangerous tasks
- Inaccessible areas (Any 2 × 1) (2)

8.4
- Robots
- Fuzzy logic
- Expert system
- Neural networks (4)

8.5
- Knowledge base
- Working storage
- Interface engine
- User interface (4)

[16]

TOTAL SECTION B: 160
GRAND TOTAL: 200

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